IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

MSc and EEE/EIE PART III/IV: MEng, BEng and ACGI

MATHEMATICS FOR SIGNALS AND SYSTEMS

Wednesday, 4 December 10:00 am

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : P. Dragotti
Second Marker(s) : S. Parbhoo

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Mathematics for Signals and Systems

There are THREE questions in this paper. Answer ALL questions. All questions carry equal marks.

Time allowed 3 hours.

Special Information for the Invigilators: none

Information for Candidates:

For questions involving matrix manipulations or calculations, please provide answers showing intermediate calculations steps.

Given a matrix $\mathbf{A}$ with full column rank, the orthogonal projection onto the space spanned by the columns of $\mathbf{A}$ is given by: $\mathbf{P} = \mathbf{A}(\mathbf{A}^H\mathbf{A})^{-1}\mathbf{A}^H$.

Consider the space $\mathbb{R}^n$ and a subspace $V \in \mathbb{R}^n$, the subspace of all vectors in $\mathbb{R}^n$ orthogonal to every vector in V is called the orthogonal complement of V and is denoted as $V^\perp$ ($V \oplus V^\perp = \mathbb{R}^n$).

The Questions

1. (a) i. Find a basis for the space of all vectors in $\mathbb{R}^4$ with $x_1 + x_2 + x_3 = 0$ and $x_3 + x_4 = 0$. [2]

   ii. Find a matrix with that subspace as its nullspace. [2]

   iii. Find a matrix with that subspace as its column space. [2]

   (b) Let $\mathbf{A} = \begin{bmatrix} 3 & 1 & -1 \end{bmatrix}$ and let V be the nullspace of $\mathbf{A}$.

   i. Find a basis for V. [2]

   ii. Find a basis for $V^{\perp}$. [2]

   iii. Find the projection matrix $\mathbf{P}_1$ that projects vectors in $\mathbb{R}^3$ onto $V^{\perp}$. [3]

   iv. Find the projection matrix $\mathbf{P}_2$ that projects vectors in $\mathbb{R}^3$ onto V. [3]

   (c) Determine the four fundamental subspaces of

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}.$$

[4]

2. (a) Suppose $\mathbf{A} \in \mathbb{R}^{10\times 12}$ has rank 8. Determine how many linearly independent solutions can be found to the homogeneous linear system $\mathbf{Ax} = \mathbf{0}$. Briefly justify your answer. [2]

(b) Suppose $\mathbf{A} \in \mathbb{R}^{8\times 10}$ has a two dimensional nullspace, show that $\mathbf{Ax} = \mathbf{b}$ can be solved for every $\mathbf{b}$. [2]

(c) Consider the following set of three measurements: (x_i, y_i): $(1, 2), (2, 1), (3, 3)$.

i. Find the line of the form $\tilde{y} = \alpha x + \beta$ that best fits (in the least-squares sense) this data. Specifically, find the line such that

$$(y_1 - \alpha x_1 - \beta)^2 + (y_2 - \alpha x_2 - \beta)^2 + (y_3 - \alpha x_3 - \beta)^2$$

is minimized. [4]

ii. Find the line of the form $\tilde{x} = \gamma y + \delta$ that best fits (in the least-squares sense) this data. [Hint: Reuse your answer from part *i*.] [4]

(d) Given

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} 1 & 2 \end{bmatrix},$$

i. find all the the possible solutions of the linear least-squares problem:

$$\min_{\mathbf{x}} \|\mathbf{Ax} - \mathbf{b}\|^2.$$

[4]

ii. For the same problem, now find the least-squares solution with the minimum norm. [4]

3. (a) Consider the matrix $\mathbf{A} = \mathbf{x}\mathbf{x}^\mathbf{T}$ where $\mathbf{x} \in \mathbb{R}^n$.

i. What is the rank of $\mathbf{A}$? Justify your answer. [3]

ii. Now assume

$$\mathbf{A} = \mathbf{x}\mathbf{x}^\mathbf{T} + \mathbf{y}\mathbf{y}^\mathbf{T}$$

where $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.

A. What is the rank of $\mathbf{A}$? Briefly justify your answer. [1]

B. Find the eigenvalues and eigenvectors of $\mathbf{A}$ when $\mathbf{x} = [1, 1]^T$ and $\mathbf{y} = [0.5, -1]^T$. [3]

(b) Consider the matrix

$$\mathbf{A} = \begin{bmatrix} 2.5 & 2.5 \\ 2.5 & 2.5 \\ 0 & 0 \end{bmatrix}.$$

You are told that the singular vector $\mathbf{u}_1$ linked to the largest singular value σ_1 is $\mathbf{u_1} = [1/\sqrt{2}, 1/\sqrt{2}, 0]^T$.

i. Calculate the singular values of $\mathbf{A}$ [3]

ii. Find the SVD of $\mathbf{A}$ [3]

iii. Find the least-square minimum norm solution to $\mathbf{A}\mathbf{x} = \mathbf{b}$ when $\mathbf{b} = [1, 0, 1]^T$. [3]

(c) Consider the Markov or "transition" matrix $\mathbf{Q} \in \mathbb{R}^{2\times 2}$. You are told that the matrix is full rank and that this Markov process has a unique steady state given by $\mathbf{u}_1 = \begin{bmatrix} 1/2, & 1/2 \end{bmatrix}^T$.

Find $\mathbf{Q}$.

(Please note that the solution might not be unique). [4]